
Session 1 (2h) — Slicer Basics: STL → G-code

Core concepts (1h)

- **What a slicer does** — takes a solid 3D model (STL/3MF) and cuts it into flat horizontal layers, then generates **G-code**: a list of exact nozzle-movement and extrusion instructions the printer executes line by line.
- **FDM basics** — Fused Deposition Modeling: a nozzle melts filament and deposits it layer by layer on a build plate. Almost all desktop/lab printers you'll use are FDM.
- **Key slicer settings you must set before every print:**
 - Printer profile (build volume, nozzle diameter)
 - Filament profile (PLA vs PETG vs ABS — different temps)
 - Layer height (0.2mm is the standard default)
- **Build plate adhesion** — first layer must stick, or the print fails immediately. Common helpers: brim, raft, bed leveling, correct bed temperature.

Hands-on practice (1h)

1. Install your slicer and select the correct printer + nozzle profile for your lab's machine.
2. Import any small STL (a test cube, or Thingiverse's classic "3DBenchy" if you want a real calibration model).
3. Slice at default settings (0.2mm layer height, 15–20% infill, PLA profile) and look at the **layer preview** — scrub through layers and watch how the shape builds up.
4. Note the estimated **print time** and **filament usage** the slicer reports — this is what you'll be asked to report on-site before starting a real print.
5. Export the G-code file (don't need to print yet — this session is about understanding the pipeline, not the machine).

Checkpoint

- ☐ Can explain STL → slicer → G-code → printer, in that order
- ☐ Correctly set printer + filament profile for your target machine
- ☐ Sliced one model and read off its estimated time/material from the preview